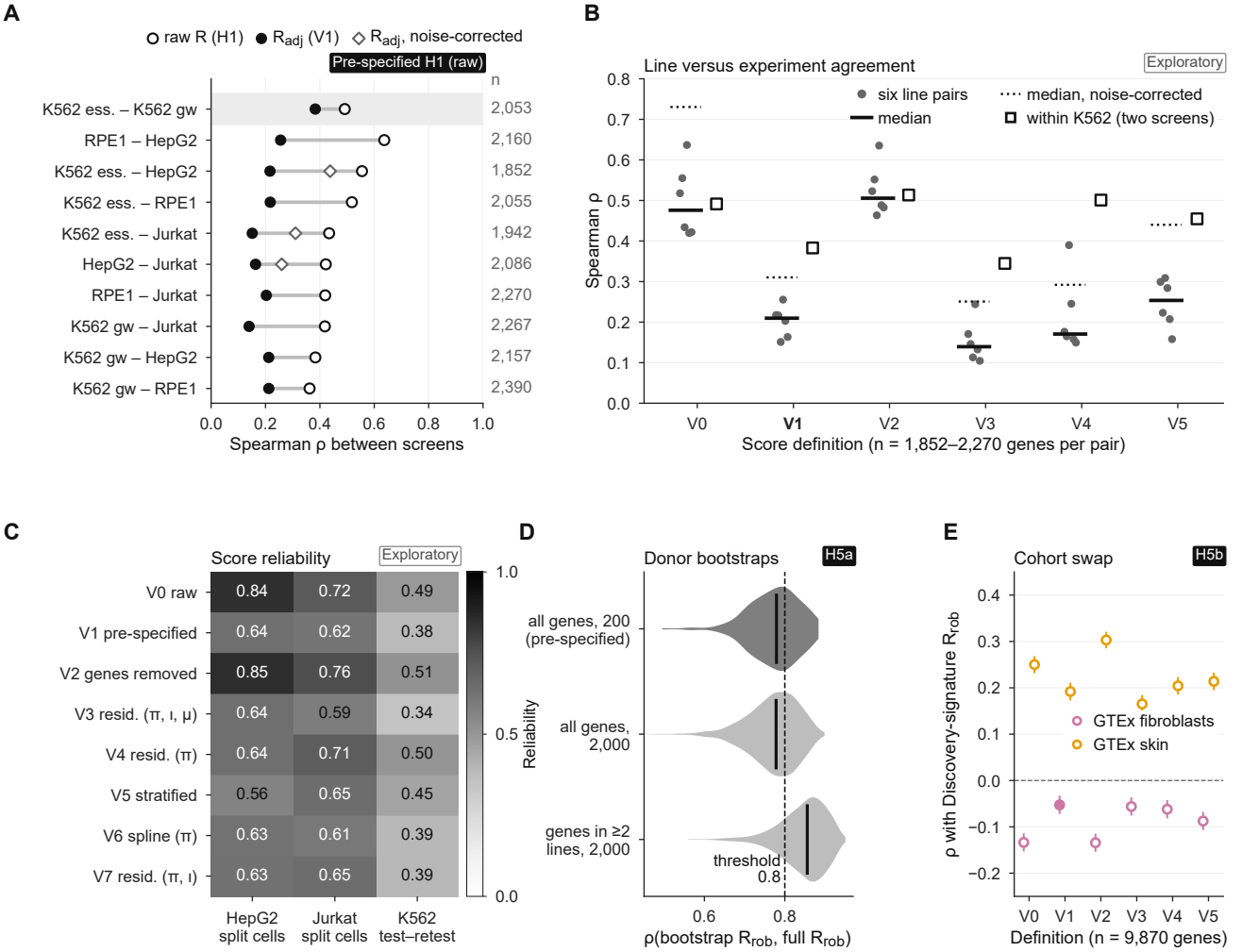

A
raw R (H1)
R_{adj} (V1)
R_{adj}, noise-corrected
Pre-specified H1 (raw)
n
K562 ess. – K562 gw
RPE1 – HepG2
K562 ess. – HepG2
K562 ess. – RPE1
K562 ess. – Jurkat
HepG2 – Jurkat
RPE1 – Jurkat
K562 gw – Jurkat
K562 gw – HepG2
K562 gw – RPE1
2,053
2,160
1,852
2,055
1,942
2,086
2,270
2,267
2,157
2,390
0.0 0.2 0.4 0.6 0.8 1.0
Spearman ρ between screens
B
Line versus experiment agreement
Exploratory
six line pairs
median
median, noise-corrected
within K562 (two screens)
Spearman ρ
0.0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8
V0 V1 V2 V3 V4 V5
Score definition (n = 1,852–2,270 genes per pair)
C
Score reliability
Exploratory
V0 raw 0.84 0.72 0.49
V1 pre-specified 0.64 0.62 0.38
V2 genes removed 0.85 0.76 0.51
V3 resid. (π, ı, μ) 0.64 0.59 0.34
V4 resid. (π) 0.64 0.71 0.50
V5 stratified 0.56 0.65 0.45
V6 spline (π) 0.63 0.61 0.39
V7 resid. (π, ı) 0.63 0.65 0.39
HepG2 split cells
Jurkat split cells
K562 test–retest
1.0
0.5
0.0
D
Donor bootstraps
H5a
all genes, 200 (pre-specified)
all genes, 2,000
genes in ≥2 lines, 2,000
Reliability
threshold 0.8
0.6 0.8
ρ(bootstrap R_{rob}, full R_{rob})
E
Cohort swap
H5b
ρ with Discovery-signature R_{rob}
0.4 0.3 0.2 0.1 0.0 −0.1 −0.2
GTEx fibroblasts
GTEx skin
V0 V1 V2 V3 V4 V5
Definition (n = 9,870 genes)